Factor (stochastic) volatility scaling and its application in portfolio optimization

Mureira & Muir (2017) with extensions to multivariate mean-variance optimization

Vienna University of Economics and Business

Bayesian Econometrics – Summer Term 2026

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Outline

1. Research Questions
2. Model Introduction
3. Moreira & Muir (2017)
4. New Methodology
5. Results

Research Questions

- **RQ 1:** Does volatility forecasting using Bayesian stochastic volatility models improve upon the Moreira & Muir (2017) historical realized volatility scaling methodology?
- **RQ 2:** Can this dynamic volatility scaling approach be effectively extended to a multi-asset allocation framework?

Stochastic Volatility in a Multivariate Framework

Goal: Model the time-varying covariance matrix Σ_t for m assets without estimating $\mathcal{O}(m^2)$ parameters per time step.

Factor Stochastic Volatility (FSV) Model

$$\mathbf{y}_t = \mathbf{\Lambda}\mathbf{f}_t + \varepsilon_t$$

- $\mathbf{\Lambda}$ $(m \times r)$: Latent factor loadings.
- $\mathbf{f}_t \sim \mathcal{N}_r(\mathbf{0}, V_t)$: r latent factors, where V_t contains independent SV processes.
- $\varepsilon_t \sim \mathcal{N}_m(\mathbf{0}, U_t)$: Idiosyncratic errors, where U_t contains independent SV processes.

Covariance Formula

$\Sigma_t = \mathbf{\Lambda} V_t \mathbf{\Lambda}' + U_t$. This isolates risk into manageable factors, creating a predictive Σ_{t+1} using only data up to time t.

FSV Intuition: What Is $\mathbf{y}_t$ and Why Factors?

Observed data: $\mathbf{y}_t$ is the vector of demeaned daily returns for all assets at time t.

Signal decomposition

$$\mathbf{y}_t = \mathbf{\Lambda}\mathbf{f}_t + \varepsilon_t$$

- $\mathbf{f}_t$: a few **common market drivers** (equity/rates/risk appetite) with time-varying volatility.
- ε_t: **asset-specific noise** with its own time-varying volatility.
- $\mathbf{\Lambda}$ maps each asset to each latent driver, so co-movement is explained by shared factors.
- Intuition: instead of estimating every covariance element directly, FSV learns a low-dimensional risk engine and updates it through time.

Why this helps volatility scaling

We get a forward-looking covariance forecast $\hat{\Sigma}_{t+1|t}$ that reacts to changing regimes and cross-asset linkages, not just past window averages.

Volatility-Managed Portfolios (Moreira & Muir, 2017)

- **Core Insight:** Volatility is highly forecastable, but expected returns are not. When volatility spikes, returns do not increase proportionally to compensate for the risk.
- **Methodology:** Scale portfolio exposure inversely proportional to recent realized variance ($\hat{\sigma}_t^2$).
- **The Scaling Rule:**

$$w_t = \frac{c}{\hat{\sigma}_t^2}$$

  where c is a constant chosen so that the unconditional average managed exposure $\bar{w}$ revolves around 1.
- **Conclusion in Paper:** This strategy yields significant alpha, increases Sharpe ratios, and sidesteps crashes, albeit relying on backwards-looking 22-day rolling realized variance.

The Proposed Methodology

Motivation: If volatility scaling derives its alpha from the predictability of variance, then using a theoretically sound predictive SV model should outperform a naive rolling-window metric.

- **Single-Asset (SPY):** Keep the Moreira & Muir scaling rule and only swap the variance input:
  - Naïve model: past 22-day realised variance
  - FSV model: 22-day horizon-average forecast variance
- **Multi-Asset Framework:**
  - Predict covariance using either rolling 22-day sample covariance or FSV one-step-ahead covariance.
  - Estimate expected returns ($\hat{\mu}_t$) using an expanding mean (causal).
  - Apply long-only mean-variance allocation within Equities, Bonds, and Gold under fixed budgets (45/45/10 and 60/40/0).

Part 1 Methodology: Exact Weight Construction

Rebalancing schedule: every 22 trading days, out-of-sample (test) period only.

Variance inputs at decision time t:

$$\hat{v}_t^{\text{naive}} = \frac{1}{22} \sum_{k=1}^{22} r_{t-k}^2$$

$$\hat{v}_t^{\text{fsv}} = \frac{1}{22} \sum_{h=1}^{22} \hat{\Sigma}^{\text{FSV}}_{t+h|t,\text{SPY},\text{SPY}}$$

Moreira & Muir scaling constants (training sample):

$$c = \left(\frac{1}{N_{\text{train}}} \sum_{t \in \text{train}} \frac{1}{\hat{v}_t} \right)^{-1}$$

Managed SPY weights:

$$w_t = \min\left(\max\left(\frac{c}{\hat{v}_t}, 0 \right), w_{\max} \right), \quad w_{\max} = 2$$

Benchmark strategy uses $w_t = 1$.

Results: Overview

Test Period

All results cover **January 2019 – December 2024** (out-of-sample). The FSV model uses expanding-window refits with **no look-ahead bias**.

Part 1 – Single Asset (SPY)

- FSV volatility scaling **dominates naïve M&M**: lower realised volatility, smaller drawdown, higher return/vol ratio, and **30% less turnover**.
- Neither scaling strategy beats the buy-and-hold Sharpe during the low-volatility, high-momentum 2019–2024 regime.

Part 1 – SPY: Volatility Forecast Quality

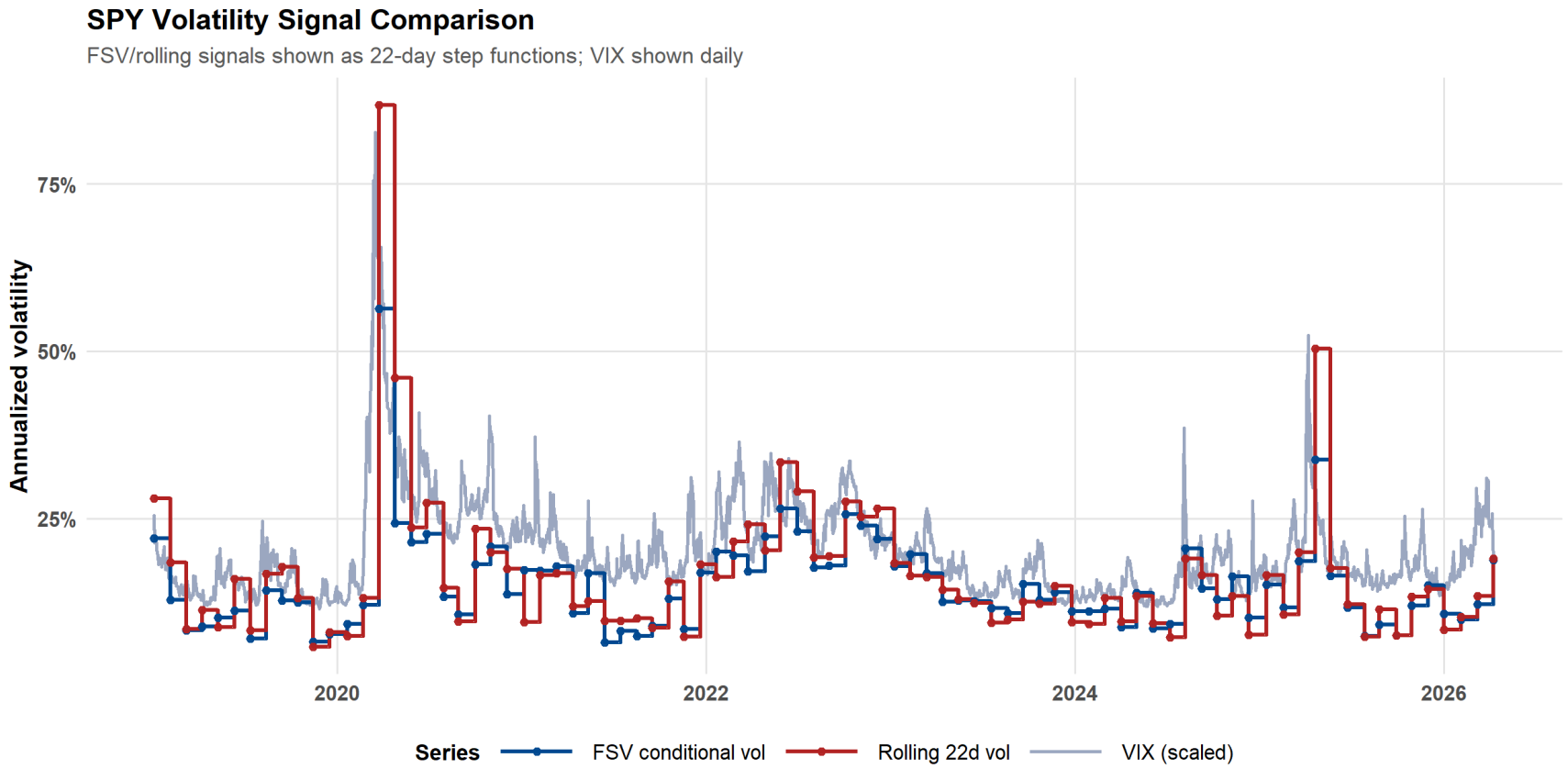


Figure 1: FSV conditional volatility vs. rolling 22-day realised volatility and VIX (annualised, test period Jan 2019 – Dec 2024). AR(1) persistence $\hat{\phi} \approx 0.999$.

Part 1 – SPY: VIX Pairwise Comparison and Deviation

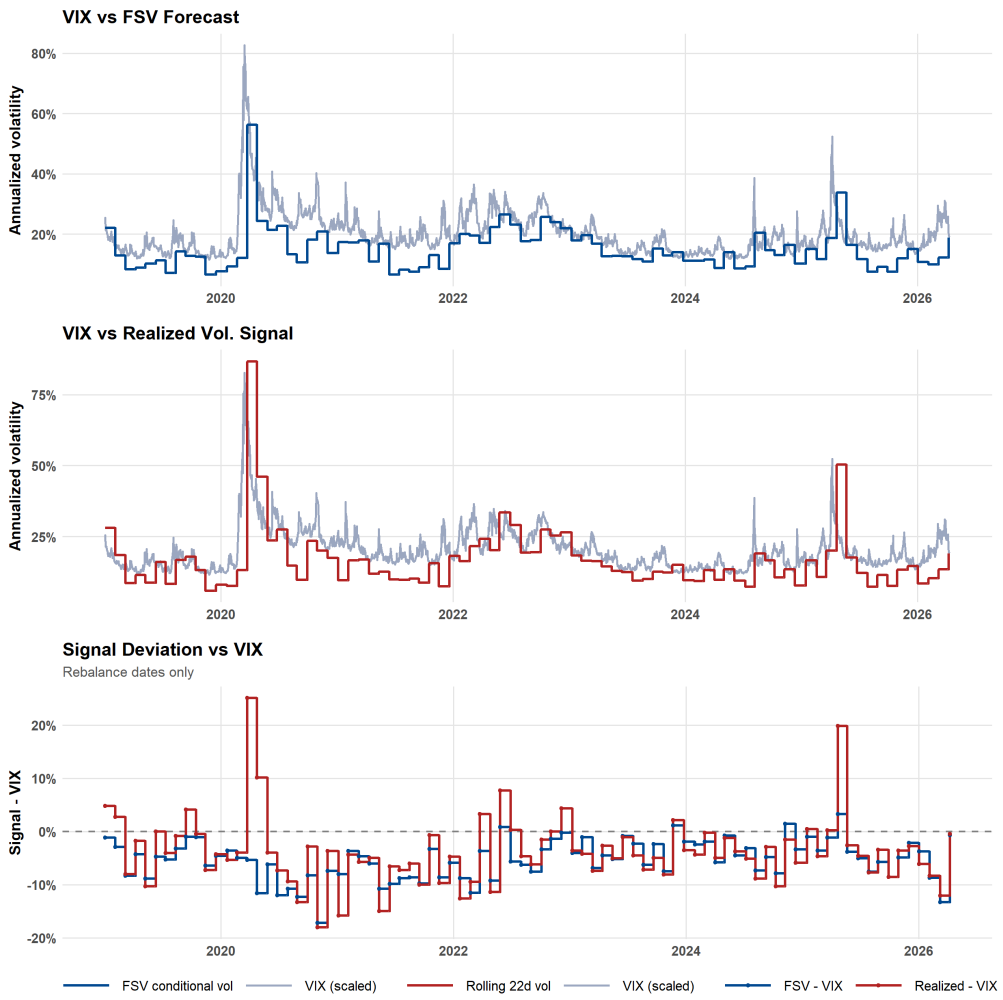


Figure 2: Top: VIX vs. FSV forecast signal. Middle: VIX vs. realized-vol signal. Bottom: deviations (signal minus VIX), evaluated on rebalance dates.

Part 1 – SPY: Cumulative Return

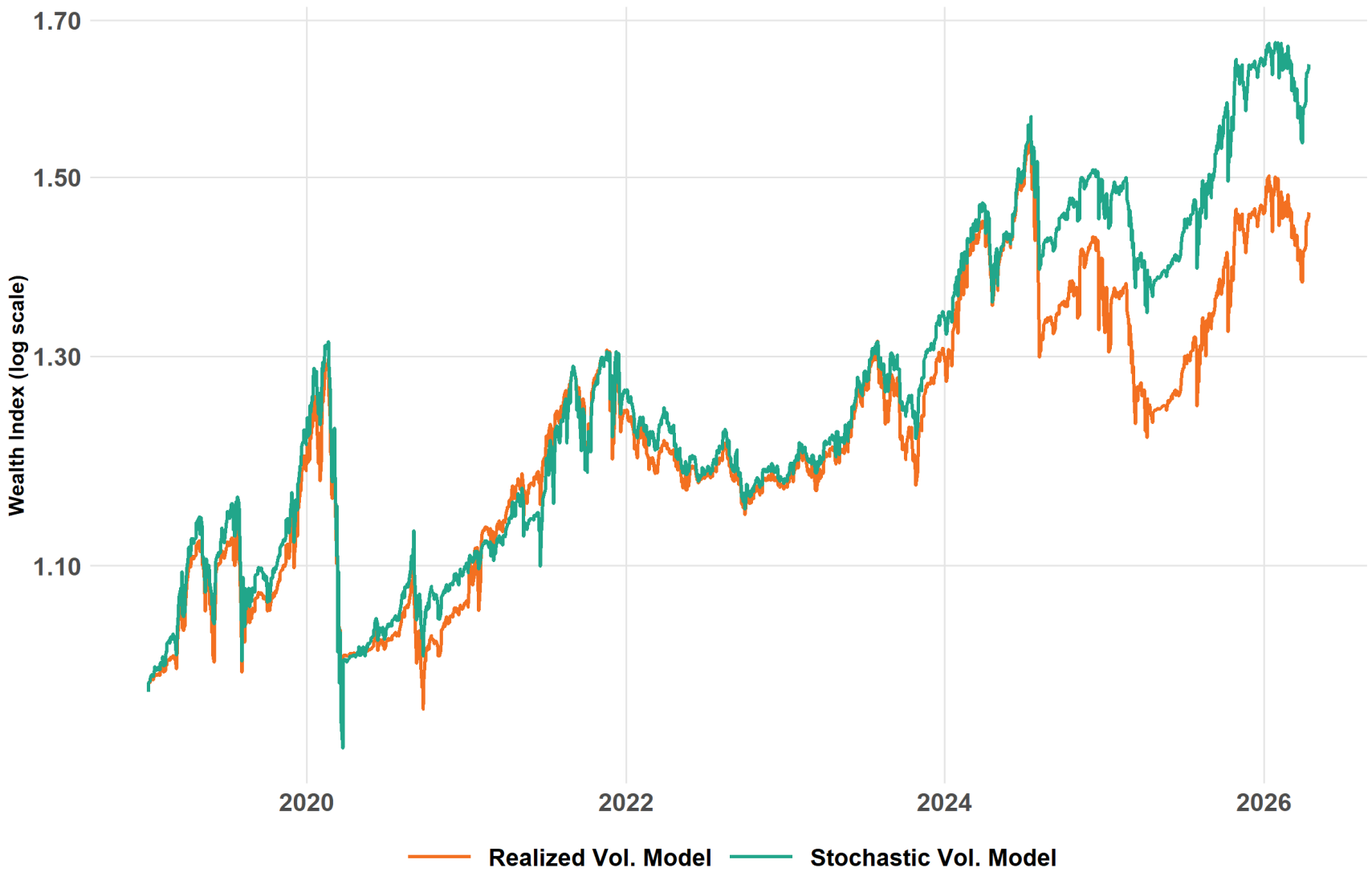


Figure 3: Cumulative wealth index (log scale), Jan 2019 – Dec 2024, 0 bps transaction costs, shown for managed models only. FSV Ret/Vol 0.496 vs. Realized Vol.

Part 1 – Results (0 vs 5 bps)

0 bps transaction cost

Strategy	Return	Vol	Ret/Vol	MaxDD	Turnover
SPY Benchmark	16.8%	19.6%	0.859	−33.7%	0.00
Realized Vol. Model	5.3%	15.2%	0.352	−25.8%	5.96
FSV	7.1%	14.3%	0.496	−27.7%	4.56

5 bps transaction cost

Strategy	Return	Vol	Ret/Vol	MaxDD	Turnover
SPY Benchmark	16.8%	19.6%	0.859	−33.7%	0.00
Realized Vol. Model	5.0%	15.2%	0.332	−25.9%	5.96
FSV	6.8%	14.3%	0.479	−27.7%	4.56

Part 2 Methodology: Exact Weight Construction

Covariance forecast at rebalance time t:

- Realized-vol model: rolling 22-day sample covariance, $\hat{\Sigma}_t^{\text{roll}}$
- Stochastic-vol model: FSV one-step-ahead covariance, $\hat{\Sigma}_{t+1|t}^{\text{fsv}}$

Expected return estimate (causal):

$$\hat{\mu}_t = \frac{1}{t} \sum_{s=1}^{t} r_s$$

Within each bucket (equity, bonds, gold), solve long-only mean-variance:

$$\max_w \; w^\top \hat{\mu}_t - \frac{\lambda}{2} w^\top \hat{\Sigma}_t w$$

subject to

$$w_i \geq 0, \qquad \sum_i w_i = 1, \qquad \lambda = 10$$

implemented via $w = \mathrm{softmax}(z)$ with optimizer bounds $z_j \in [-20, 20]$.

Bucket solutions are scaled by fixed top-level budgets:

$$\text{pf1} : (0.45, 0.45, 0.10), \qquad \text{pf2} : (0.60, 0.40, 0.00)$$

Boundaries and Return Construction

Weight boundaries

- Part 1: scalar SPY weight constrained to $w_t \in [0, 2]$.
- Part 2: long-only simplex constraints ($w_i \geq 0$, weights sum to 1 in each bucket and overall), implemented numerically with $z_j \in [-20, 20]$.

Gross returns

$$r^{\text{gross}}_{t+1,\text{Part1}} = w_t \, r_{\text{SPY},t+1}, \qquad r^{\text{gross}}_{t+1,\text{Part2}} = w_t^\top r_{t+1}$$

Financing cost for leverage

$$\text{FC}_{t+1} = \max(0, \text{gross exposure}_t - 1) \left(r_{\text{SHY},t+1} + \frac{0.003}{252} \right)$$

with floor at zero implied by the borrowing-rate proxy.

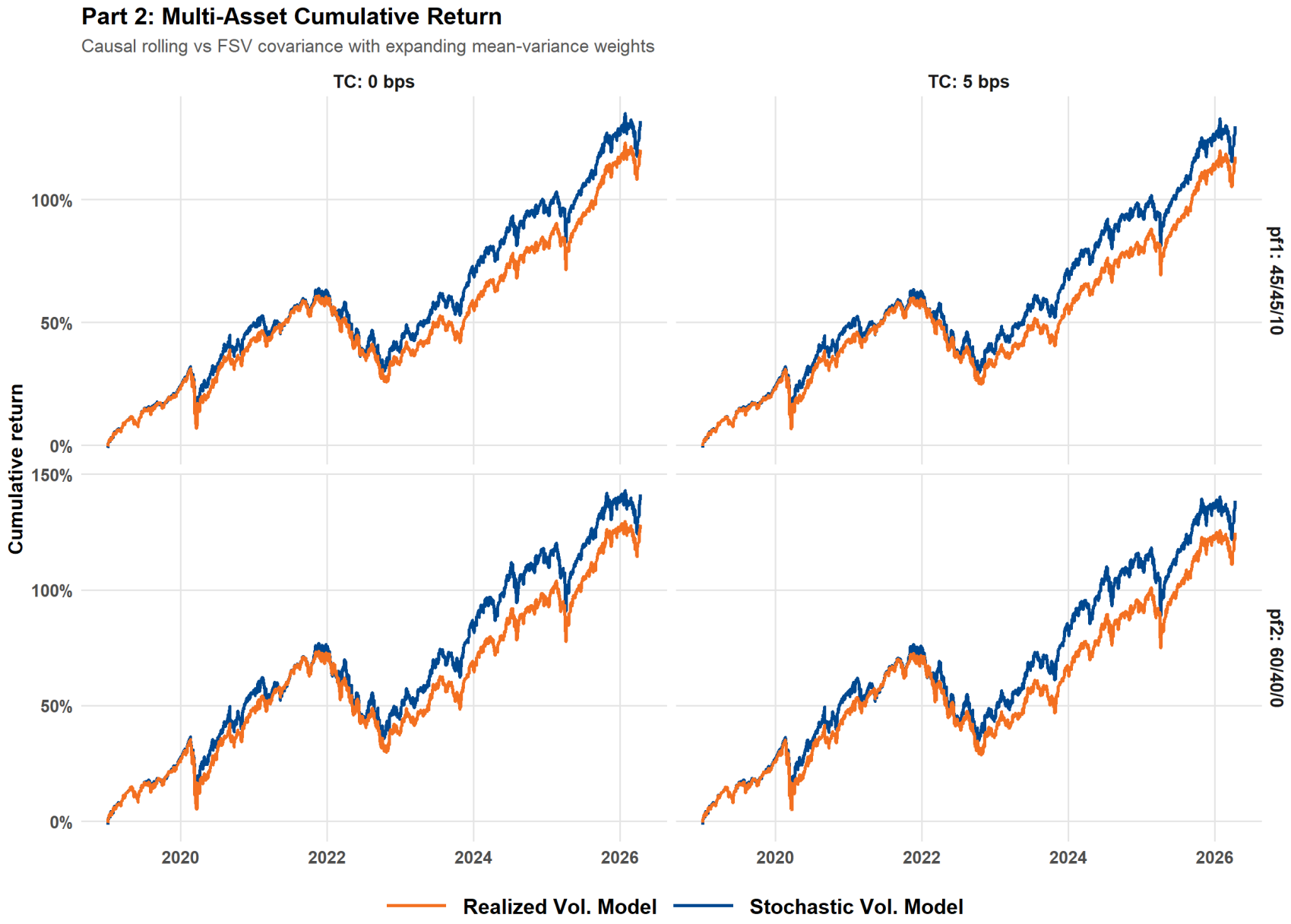
Net returns with transaction costs

$$r^{\text{net}}_{t+1} = r^{\text{gross}}_{t+1} - \text{FC}_{t+1} - \frac{\text{tcost}_{\text{bps}}}{10{,}000} \cdot \text{turnover}_t$$

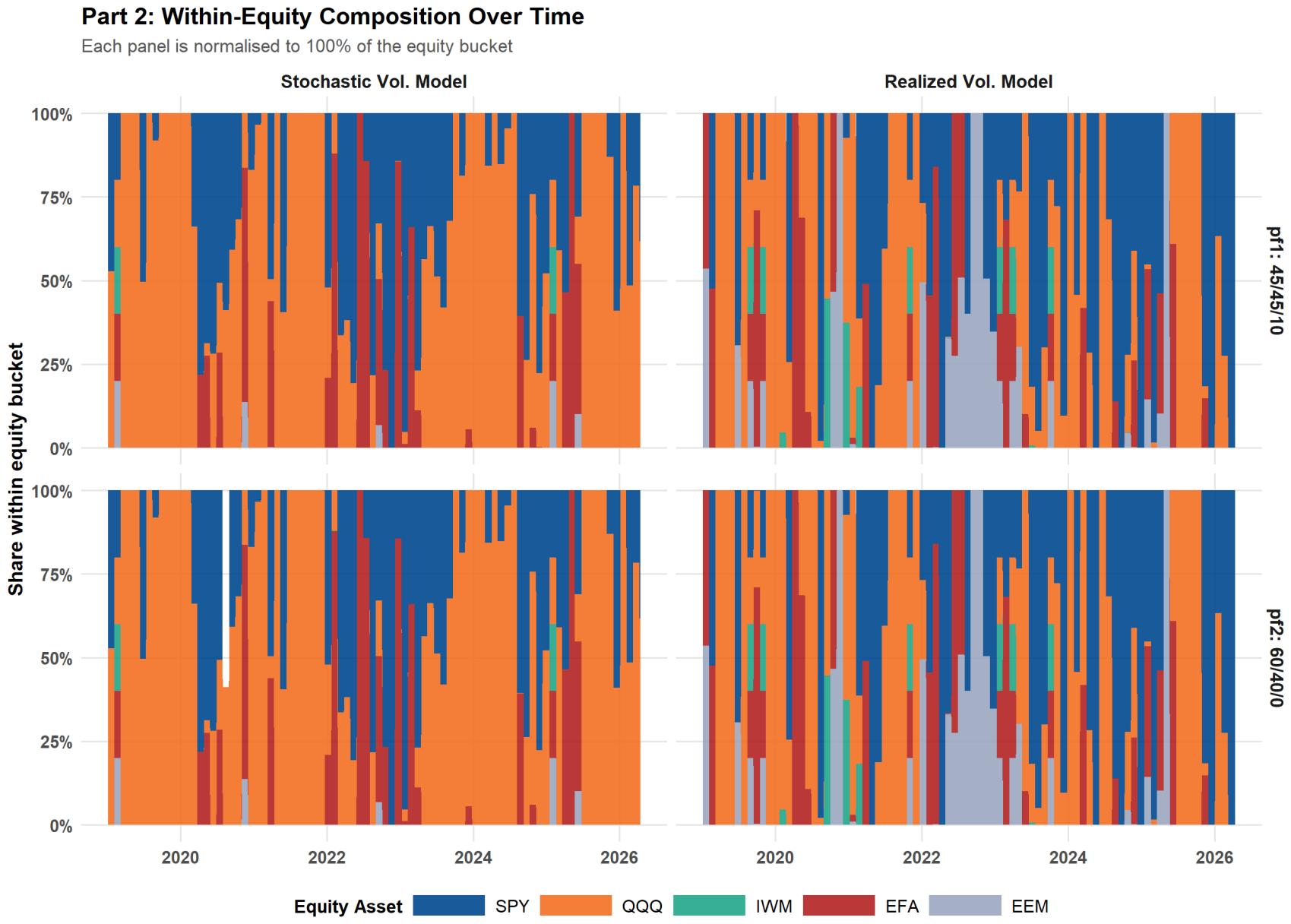
where transaction costs are evaluated at 0 and 5 bps, and one-way turnover is

$$\text{turnover}_t = \begin{cases} |w_t - w_{t-1}|, & \text{Part 1 (single weight)} \\ \frac{1}{2} \sum_i |w_{i,t} - w_{i,t-1}|, & \text{Part 2 (portfolio weights)} \end{cases}$$

Part 2 – Multi-Asset: Cumulative Return



Part 2 – Multi-Asset: Portfolio Weight Dynamics



Part 2 – Results (0 vs 5 bps)

0 bps transaction cost

Budget	Strategy	Return	Vol	Ret/Vol	MaxDD	Turnover
pf1	Realized Vol. Model	11.5%	10.5%	1.097	−21.9%	3.78
pf1	Stochastic Vol. Model	12.3%	11.3%	1.089	−20.9%	2.73
pf2	Realized Vol. Model	12.0%	12.8%	0.942	−25.3%	4.54
pf2	Stochastic Vol. Model	12.9%	13.7%	0.939	−23.5%	3.24

5 bps transaction cost

Budget	Strategy	Return	Vol	Ret/Vol	MaxDD	Turnover
pf1	Realized Vol. Model	11.3%	10.5%	1.077	−22.0%	3.78
pf1	Stochastic Vol. Model	12.1%	11.3%	1.075	−21.0%	2.73
pf2	Realized Vol. Model	11.8%	12.8%	0.922	−25.5%	4.54
pf2	Stochastic Vol. Model	12.7%	13.7%	0.925	−23.7%	3.24

Discussion & Caveats

Main Findings

- **RQ 1 (Single Asset):** FSV scaling *improves upon* naïve M&M — better risk-adjusted returns, lower drawdown, and lower turnover. However, neither beats buy-and-hold in the 2019–2024 low-VIX regime, consistent with M&M's own observation that alpha is regime-dependent.
- **RQ 2 (Multi-Asset):** FSV covariance forecasts yield a *practically superior* portfolio on return, drawdown, and turnover at both budgets, with the largest advantage being **transaction-cost efficiency**.

Key Caveats & Deviations

- **Test window:** Jan 2019 onward; GFC and pre-2019 crises are *not* in the evaluation period.
- **Leverage:** Scaling constant c is calibrated on the full training window; realised exposures are dynamic and occasionally hit the $2\times$ cap (mean exposure: Realized Vol. Model $0.80\times$, FSV $0.78\times$).
- **AR(1) SV estimation:** Parameters estimated by OLS on posterior-mean log-variance paths for tractability.

Appendix – FSV Volatility: Local vs GCP

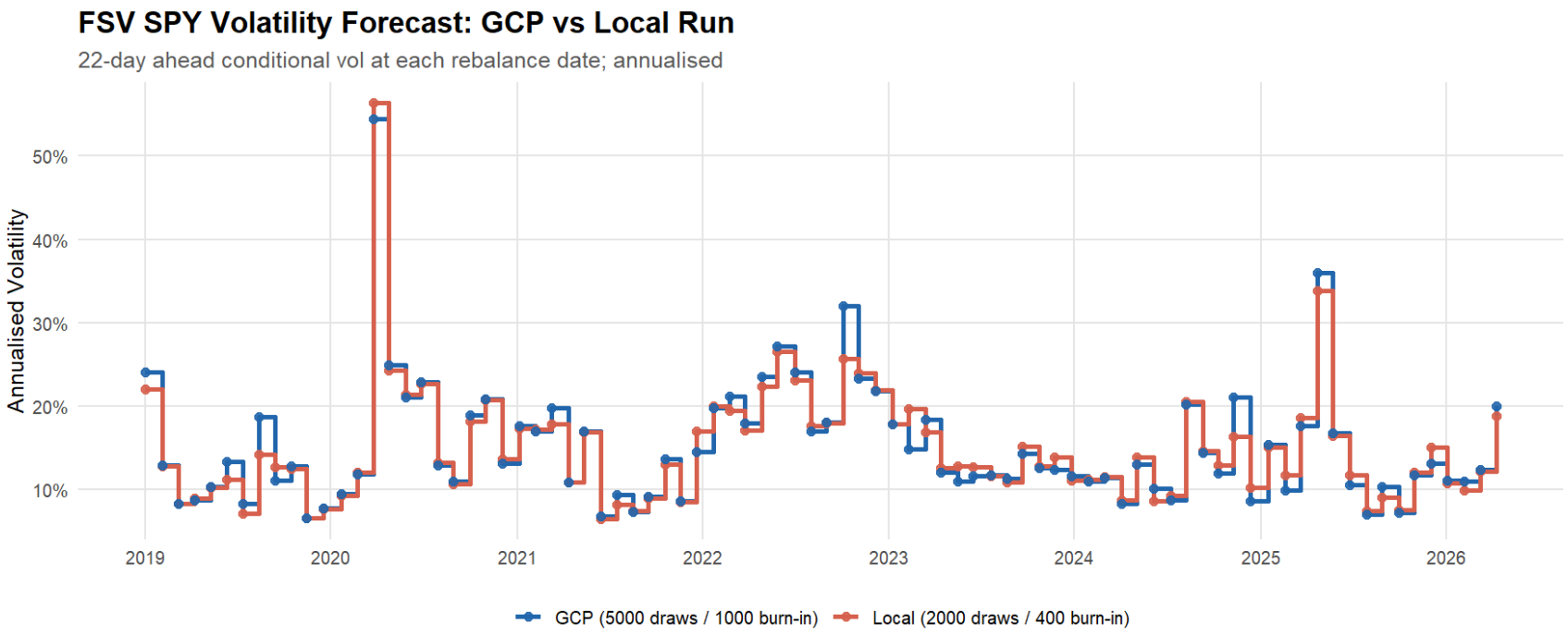
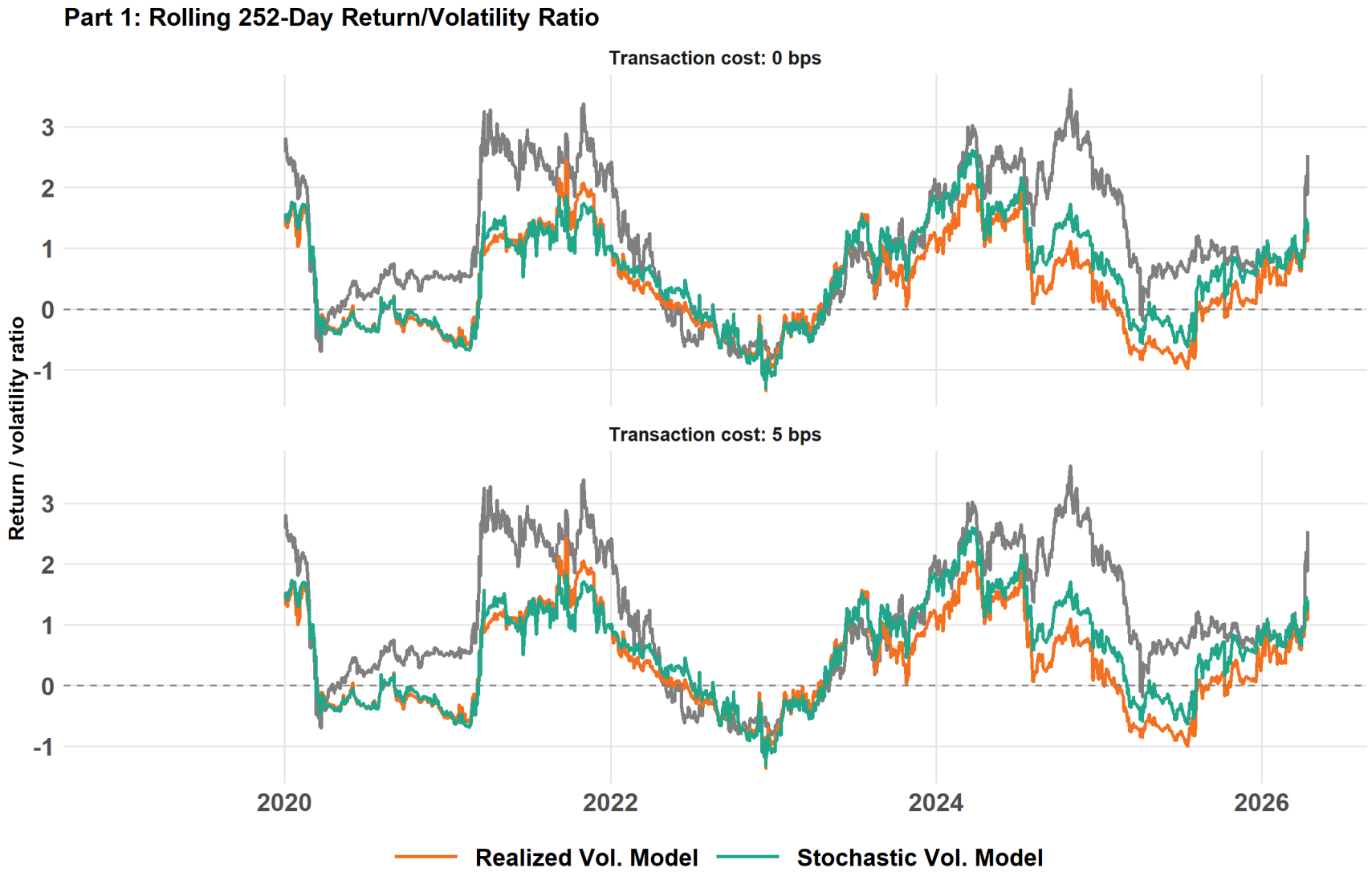
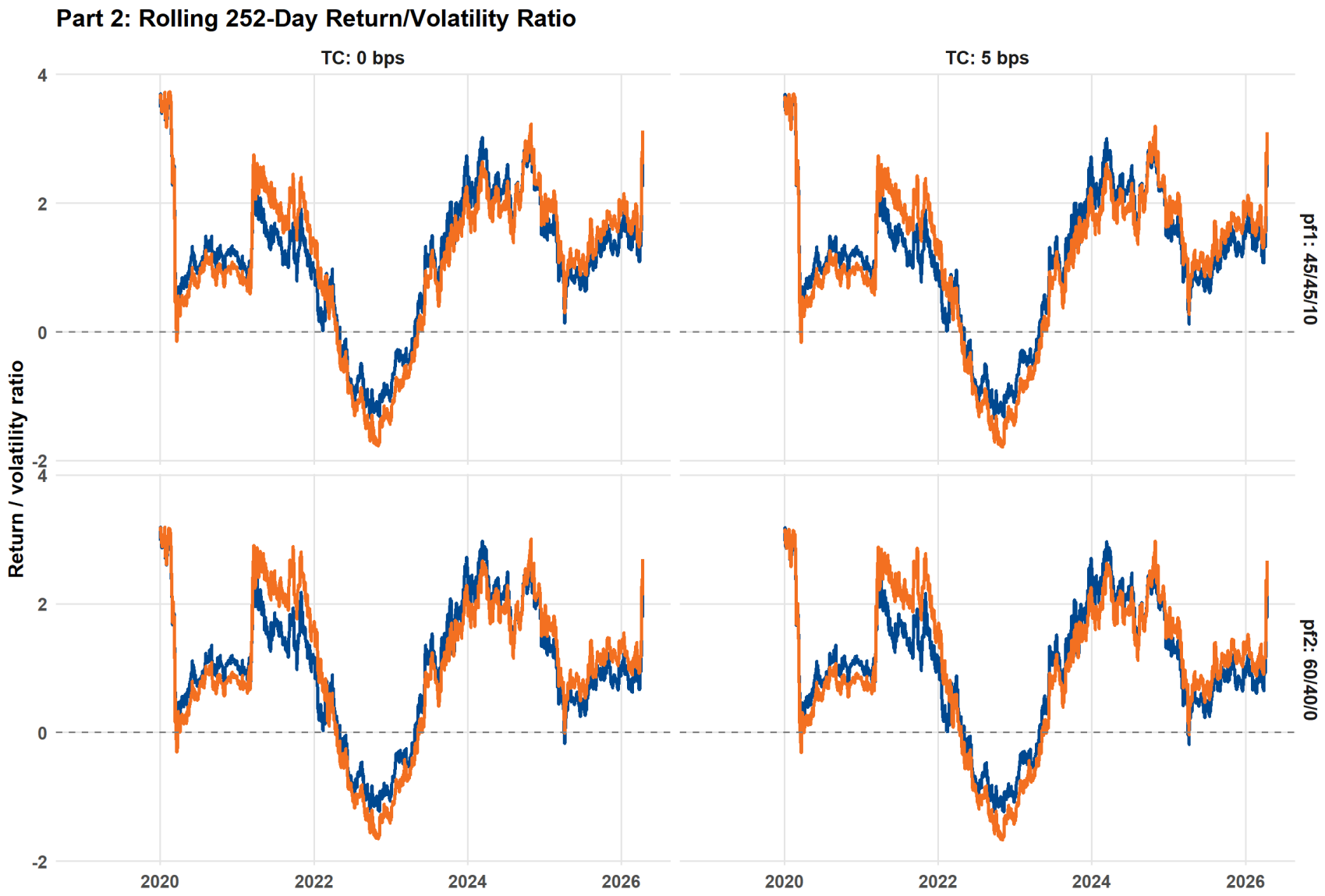


Figure 6: Comparison of Part 1 FSV volatility signal under local vs GCP MCMC settings.

Appendix – Part 1: Rolling 252-Day Return/Volatility Ratio



Appendix – Part 2: Rolling 252-Day Return/Volatility Ratio



Appendix – Global Minimum-Variance: Cumulative Return

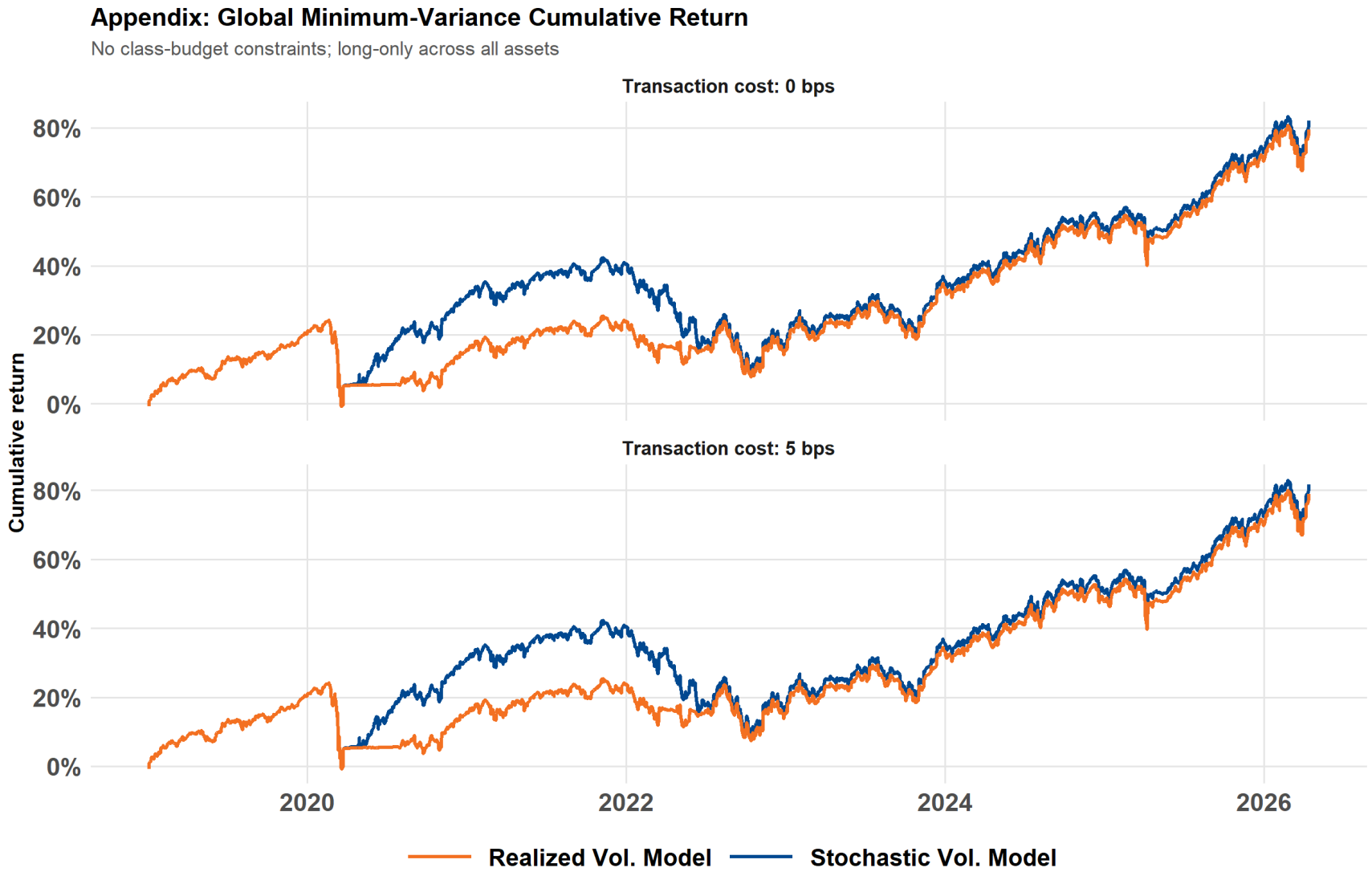
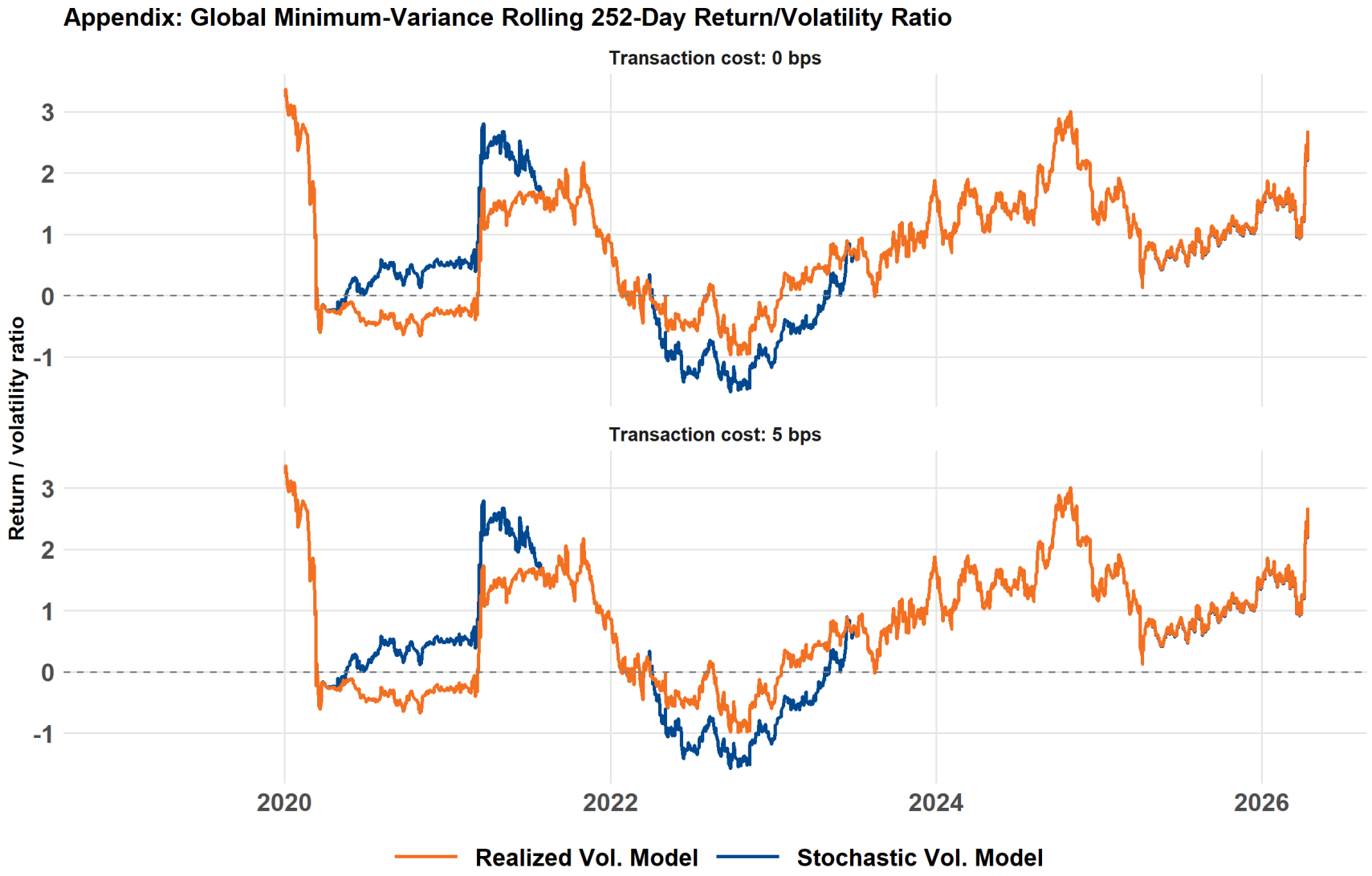


Figure 9: Global long-only minimum-variance portfolios (no class-budget constraints): realized-vol covariance vs. stochastic-vol covariance, shown for 0 and 5 bps costs.

Appendix – Global Minimum-Variance: Rolling Return/Volatility Ratio



Appendix – Full Performance Metrics

Part 1: Single-Asset SPY (0 bps, Jan 2019 – Dec 2024)

	SPY	Realized Vol.	FSV
Ann. Return	16.8%	5.3%	7.1%
Ann. Vol	19.6%	15.2%	14.3%
Ret/Vol	0.859	0.352	0.496
Max Drawdown	−33.7%	−25.8%	−27.7%
ES (5%)	−2.95%	−2.60%	−2.32%
Turnover (p.a.)	0.00	5.96	4.56

Part 2: Multi-Asset Budgets (0 bps)

Portfolio	Strategy	Ret	Vol	R/V	MaxDD	Turn.
pf1 (45/45/10)	Realized Vol. Model	11.5%	10.5%	1.097	−21.9%	3.78
pf1 (45/45/10)	FSV	12.3%	11.3%	1.089	−20.9%	2.73
pf2 (60/40/0)	Realized Vol. Model	12.0%	12.8%	0.942	−25.3%	4.54
pf2 (60/40/0)	FSV	12.9%	13.7%	0.939	−23.5%	3.24

Appendix – Asset Universe and Class Weights

Ticker	Name	Class	Within-Class	pf1 Class	pf2 Class
SPY	SPDR S&P 500 ETF Trust	Equity	40%	45%	60%
QQQ	Invesco QQQ Trust	Equity	20%	45%	60%
IWM	iShares Russell 2000 ETF	Equity	20%	45%	60%
EFA	iShares MSCI EAFE ETF	Equity	10%	45%	60%
EEM	iShares MSCI Emerging Markets ETF	Equity	10%	45%	60%
TLT	iShares 20+ Year Treasury Bond ETF	Bonds	30%	45%	40%
SHY	iShares 1–3 Year Treasury Bond ETF	Bonds	30%	45%	40%
LQD	iShares iBoxx $ Inv. Grade Corp. Bond ETF	Bonds	25%	45%	40%
HYG	iShares iBoxx $ High Yield Corp. Bond ETF	Bonds	15%	45%	40%
GLD	SPDR Gold Shares	Gold	100%	10%	0%

Data source: Yahoo Finance via `quantmod::getSymbols()`, daily adjusted prices.

Appendix – Local vs GCP Differences (Part 1)

Strategy	Return	Vol	Sharpe	Max DD	Turnover
SPY Benchmark	16.9%	19.6%	0.865	−33.7%	0.00
Naive MM (Realized)	5.4%	15.1%	0.355	−25.8%	5.95
FSV MM – Local	7.1%	14.3%	0.496	−27.7%	4.56
FSV MM – GCP	5.8%	15.8%	0.365	−31.0%	5.37

Key difference: with deeper GCP convergence, FSV and Naive are nearly indistinguishable on Sharpe (0.365 vs 0.355).

Appendix – Local vs GCP Differences (Part 2)

Strategy (pf1, 0 bps)	Return	Vol	Sharpe	Max DD	Turnover
Rolling MV – Local	11.5%	10.5%	1.097	−21.9%	3.78
Rolling MV – GCP	11.5%	10.5%	1.099	−21.9%	3.78
FSV MV – Local	12.3%	11.3%	1.089	−20.9%	2.73
FSV MV – GCP	12.1%	11.2%	1.079	−21.6%	2.91

Multi-asset conclusion is stable across compute settings; differences are small and mostly in the FSV leg.

Appendix – FSV Priors Used

Loadings Priors

`priorfacloadtype="rowwiseng"`, `priorfacload=0.1`, `priorng=c(1,1)`: row-wise Normal-Gamma shrinkage, pulling small loadings toward zero while allowing large loadings when supported by data.

SV AR(1) Priors

`priormu=c(0,10)` (diffuse on μ), `priorphiidi=priorphifac=c(10,3)` (persistent ϕ, prior mean $10/13 \approx 0.77$), `priorsigmaidi=priorsigmafac=1` (Exponential(1) on vol-of-vol), and `priorh0idi=priorh0fac="stationary"` for initial states.

Key Implication in This Project

Posterior persistence moved close to the upper clamp in many series (around 0.999), indicating that the data dominates the prior on ϕ and implies very strong volatility persistence, which is typical in financial return volatility.

AI Disclaimer

AI-Assisted Tools Used

The following AI-assisted language and coding models were used in the preparation of this work: **Claude** (Sonnet 4.6 and Opus 4.6, Anthropic), **ChatGPT Codex** (OpenAI), and **Gemini** (Google).

- **Claude** and **ChatGPT Codex** were used to support the implementation of the analytical pipeline, including code generation and debugging, validation of model outputs, summarisation of results, and the production of charts and figures.
- **Gemini** was used to cross-validate selected results and to assist with spelling and language checking.

All AI-generated outputs were reviewed, critically assessed, and adapted by the authors.